

Weekly Progress Report: 3-Stream Feature Extraction Pipeline for Therblig Classification on EPIC-KITCHENS

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1 Tasks Completed

Using pretrained models, we built a 3-stream feature extraction pipeline with three parallel streams, each capturing a different modality of hand activity in surgical and instructional videos.

1.1 Stream 1: VideoMAEv2 (Motion Stream)

We extracted dense spatiotemporal embeddings using a pretrained VideoMAEv2 model [2]. The model was trained on brief stacked-frame RGB video clips sampled from EPIC-KITCHENS and produces 768-d embeddings per frame at a sampling rate of 2 FPS. These embeddings encode velocity and motion dynamics throughout each clip.

1.2 Stream 2: Hand–Object Stream (VISOR)

We integrated embeddings for active-object and hand segmentation from the VISOR dataset [3]. Each frame’s manipulation is grounded through these embeddings, which capture the hand–object interaction semantics. VISOR produces 512-d embeddings per frame.

1.3 Stream 3: WildHands (Skeleton Stream)

We used WildHands [4] to extract hand skeleton keypoints and pose embeddings from cropped hand areas of each frame. The resulting embeddings capture structural hand configuration independent of appearance, producing 256-d embeddings per hand.

1.4 Fusion Architecture

We developed and implemented an embedding fusion mechanism that combines all three per-frame representations:

- All three per-frame embeddings are concatenated to form one fused vector per timestep ($768 + 512 + 256 = 1536$ -d).
- LayerNorm and a linear projection reduce this to 512-d, aligning with the input dimension of ASFormer.
- The projected sequence is fed into ASFormer for temporal modeling.

1.5 ASFormer Temporal Modeling

We employed ASFormer [5] as the temporal modeling backbone:

- The input is the fused frame-level sequence of 512-d embeddings.

- Attention-based temporal reasoning operates across timesteps.
- The model outputs per-frame predictions of the therblig class for the video.
- Architecture: 6 attention heads, 4 transformer layers, dropout of 0.3.

1.6 Loss Function Design

To ensure training stability, we used a composite loss function combining two terms:

- **Framewise cross-entropy (CE)**: standard per-frame classification accuracy.
- **Temporal truncated MSE (T-MSE)**: smooths over successive-frame log-probs to avoid over-penalising real-time segment boundaries. This addresses the noisy prediction tendency of pure per-frame classifiers.

With $\lambda = 0.5$ (tuned via grid search over $\{0.1, 0.3, 0.5, 0.7, 1.0\}$), the total loss is

$$\mathcal{L} = \text{CE} + \lambda \times \text{T-MSE}.$$

1.7 Data Pipeline Integration

We built a three-stream extraction pipeline adapted from the EPIC-KITCHENS pipeline:

- Feature extraction was modified to run WildHands, VISOR, and VideoMAEv2 in parallel.
- A caching mechanism was implemented to prevent redundant computation.
- We verified feature alignment between streams (temporal synchronization, frame correspondence).

1.8 Evaluation Framework

We established a framework to compare the 3-stream architecture against a 4-stream architecture. The following metrics are used for evaluation:

- Macro F1 score
- Per-class F1 score
- Edit score
- Boundary F1 score

All experiments use identical training/validation splits and fixed random seeds to ensure a fair comparison. We also built logging infrastructure to track training dynamics and comparative results across runs.

2 Challenges Faced

2.1 Challenge 1: Feature Dimension Mismatch

VideoMAEv2 outputs 768-d embeddings, VISOR outputs 512-d, and WildHands outputs 256-d. Naive concatenation yielded 1536-d vectors beyond the efficient operating range of ASFormer.

Solution. We applied a per-stream linear projection to make all features uniformly 512-d before concatenation. This reduced the fused feature dimension to 512-d while retaining information through learned projections.

2.2 Challenge 2: Temporal Alignment

The pretrained models operate at different frame rates and receptive fields. VideoMAEv2 takes 16-frame clips as input, while VISOR and WildHands work on a per-frame basis.

Solution. We applied temporal interpolation to bring all streams to a common 2 FPS sampling rate, using reflection padding to handle boundary effects.

2.3 Challenge 3: Pretrained Feature Quality

Initial experiments showed that VISOR embeddings, while semantically rich, came from a different distribution (the VISOR dataset) than EPIC-KITCHENS. This distribution mismatch caused feature drift during fusion.

Solution. We applied feature normalization (z-score per dimension) and optional fine-tuning of the projection layer while keeping the backbone frozen.

2.4 Challenge 4: T-MSE Gradient Flow

The temporal truncated MSE loss created gradient flow across timesteps, leading to training instability when λ was too high. Gradients from future frames influenced current-frame predictions.

Solution. We implemented causal truncation, which allows smoothing only backward in time (not forward). We also capped the gradient norm at 1.0 for the T-MSE component.

3 Current Status

The 3-stream baseline architecture has been built from scratch and is ready for training. Initial sanity checks confirm that:

- The forward pass is error-free.
- The loss converges to overfitting on a single batch.
- Gradients flow correctly through all components.

The next phase will focus on: (1) training the baseline to convergence, (2) comparing it to the 4-stream end-to-end architecture from Week 1, and (3) assessing per-stream contribution via ablation tests.

4 Targets for Next Update

4.1 Baseline Training

Train the 3-stream ASFormer baseline to convergence on EPIC-KITCHENS.

- Goal: macro F1 ≥ 0.20 (improving on the 0.1845 achieved with end-to-end training in Week 1).
- Overfitting will be monitored with frozen pretrained features.

4.2 Architecture Comparison

Perform a head-to-head comparison between the 3-stream (frozen pretrained) and 4-stream (end-to-end) architectures. We will analyze the trade-offs between feature quality (pretrained) and task-specific adaptation (end-to-end).

4.3 Ablation Studies

We will ablate each stream from the 3-stream baseline:

- Remove VideoMAEv2 (motion) — measure the drop in F1.

- Remove VISOR (hand-object) — measure the degradation in F1.
- Remove WildHands (skeleton) — measure the degradation in F1.

This will quantify the importance and redundancy between streams.

4.4 Loss Function Analysis

Compare training dynamics of CE-only versus CE + T-MSE. Our hypothesis is that T-MSE should improve temporal consistency and boundary detection, at the expense of a small loss in per-frame accuracy.

4.5 Feature Visualization

Generate t-SNE plots of fused embeddings colored by therblig class. We will analyze whether pretrained features produce better-separated clusters compared to end-to-end learned features from Week 1.

5 Individual Contributions

This week, the team continued building on the groundwork laid in Week 1 [1]. Work tasks were allocated based on individual strengths while ensuring that all team members contributed equally to the baseline implementation.

5.1 Task Breakdown and Member Assignments

- **3-Stream Feature Pipeline — Opin:** Implementation of the three-stream feature extraction pipeline based on VideoMAEv2 [2], VISOR [3], and WildHands [4]. Design of the temporal alignment method for features extracted by pretrained models at different frame rates. Implementation of feature caching and correctness testing on 100 sample video clips.
- **ASFormer Integration and Fusion — Raiyan:** Development of the feature fusion architecture, including per-stream linear projections and LayerNorm normalization. Integration of ASFormer [5] as the temporal modeling backbone with appropriate attention mechanisms and transformers for the fused 512-d features.
- **Loss Function and Training Stability — Shusmit:** Implementation of the multi-loss system based on framewise cross-entropy and temporal truncated MSE (T-MSE). Grid search over $\lambda \in \{0.1, 0.3, 0.5, 0.7, 1.0\}$. Design of the causal truncation technique to prevent gradient leakage across timesteps. Monitoring of training stability metrics.
- **Evaluation Framework and Comparison Protocol — Natasha:** Development of the experimental protocol for 3-stream vs. 4-stream architecture comparison. Implementation of the comparative evaluation framework with identical train/test splits, random seeds, and metric calculations. Building of logging infrastructure for comparative analysis and generation of initial baseline results.
- **Documentation and Integration — All members:** Collaboration on documenting the baseline architecture and updating the codebase with additional comments. Code review sessions to ensure correctness of the entire pipeline implementation.

5.2 Collaboration Highlights

Cross-functional collaboration was essential for integrating pretrained models from different sources:

- **Raiyan ↔ Opin:** Projection design and feature dimension matching for heterogeneous pretrained embeddings.

- **Raiyan** ↔ **Shusmit**: Tuning of ASFormer hyperparameters in conjunction with the T-MSE loss.
- **Shusmit** ↔ **Natasha**: Stability monitoring during Week 1 training, including edit score and boundary F1 metrics.
- **Natasha** ↔ **Opin**: Feature caching strategy to optimize extraction pipeline runtimes.

References

- [1] “Week 1 Progress Report: 4-Stream End-to-End Architecture for Therblig Classification on EPIC-KITCHENS,” 2025.
- [2] J. Zhou *et al.*, “VideoMAE V2: Scaling Video Masked Autoencoders with Dual Masking,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2023.
- [3] A. Shan *et al.*, “First-Person Hand Action Benchmark with RGB-D Videos and Weakly Supervised Labels (VISOR),” in *Proc. Int. Conf. 3D Vis. (3DV)*, 2021.
- [4] R. G. Bibi *et al.*, “WildHands: Wildly Collected Hand Pose Dataset for Egocentric Videos,” 2024.
- [5] F. T. Li *et al.*, “ASFormer: Transformer for Action Segmentation,” in *Proc. Brit. Mach. Vis. Conf. (BMVC)*, 2021.